Healthcare Analytics SQL Reference

This table presents priority healthcare analytics questions, aligned SQL queries, and sample outputs for stakeholder review.

Executive Overview

This summary provides a structured view of key healthcare analytics questions across patient demographics, diagnoses, appointments, lab activity, risk segmentation, and readmissions. Together, these analyses support operational review, population health monitoring, and data-informed decision-making by aligning each business question with its corresponding SQL logic and sample output.

<table>
<tr><th>Category and Business Question</th><th>Microsoft Server SQL Query</th><th>Result Set</th></tr>
<tr>
<td>Demographics and Diagnoses
What is the patient profile by age and gender?</td>
<td>SELECT
gender,
CASE
WHEN DATEDIFF(year, date_of_birth, GETDATE()) BETWEEN 0 AND 17 THEN 'Pediatric'
WHEN DATEDIFF(year, date_of_birth, GETDATE()) BETWEEN 18 AND 64 THEN 'Adult'
ELSE 'Senior'
END AS age_group,
COUNT(*) AS patient_count
FROM [Healthcare_Database].[dbo].Patients
GROUP BY
gender,
CASE
WHEN DATEDIFF(year, date_of_birth, GETDATE()) BETWEEN 0 AND 17 THEN 'Pediatric'
WHEN DATEDIFF(year, date_of_birth, GETDATE()) BETWEEN 18 AND 64 THEN 'Adult'
ELSE 'Senior'
END
ORDER BY patient_count Desc</td>
<td>
<table>
<tr><th></th><th>gender</th><th>age_group</th><th>patient_count</th></tr>
<tr><td>1</td><td>Female</td><td>Adult</td><td>44</td></tr>
<tr><td>2</td><td>Male</td><td>Adult</td><td>26</td></tr>
<tr><td>3</td><td>Male</td><td>Senior</td><td>14</td></tr>
<tr><td>4</td><td>Female</td><td>Senior</td><td>7</td></tr>
<tr><td>5</td><td>Female</td><td>Pediatric</td><td>6</td></tr>
<tr><td>6</td><td>Male</td><td>Pediatric</td><td>3</td></tr>
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</td>
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</table>

Demographics and Diagnoses

How do diagnoses differ by age group and gender?

```
SELECT
gender,
diagnosis,
CASE
WHEN DATEDIFF(year, date_of_birth, GETDATE()) BETWEEN 0 AND 17 THEN 'Pediatric'
WHEN DATEDIFF(year, date_of_birth, GETDATE()) BETWEEN 18 AND 64 THEN 'Adult'
ELSE 'Senior'
END AS age_group,
COUNT(*) AS patient_count
FROM [Healthcare_Database].[dbo].Patients AS p
INNER JOIN [Healthcare_Database].[dbo].[Outpatient Visits] AS ov
ON p.patient_id = ov.patient_id
WHERE diagnosis <> 'Unknown'
GROUP BY
gender,
diagnosis,
CASE
WHEN DATEDIFF(year, date_of_birth, GETDATE()) BETWEEN 0 AND 17 THEN 'Pediatric'
WHEN DATEDIFF(year, date_of_birth, GETDATE()) BETWEEN 18 AND 64 THEN 'Adult'
ELSE 'Senior'
END
ORDER BY patient_count DESC, diagnosis ASC
```

	gender	diagnosis	age_group	patient_count
1	Female	Diabetes	Adult	20
2	Female	Hypertension	Adult	13
3	Female	Muscle Injury	Adult	12
4	Female	Respiratory Illness	Adult	12
5	Male	Diabetes	Adult	11
6	Male	Hypertension	Adult	9
7	Female	Hyperlipidemia	Adult	7
8	Male	Muscle Injury	Adult	6
9	Male	Respiratory Illness	Senior	6
10	Female	Migraine	Adult	5
11	Male	Allergic Reaction	Adult	4
12	Male	Common Cold	Adult	4
13	Male	Diabetes	Senior	4
14	Female	Ear Infection	Pediatric	4
15	Male	Muscle Injury	Senior	4
16	Male	Respiratory Illness	Adult	4
17	Female	Respiratory Illness	Senior	4
18	Male	Bronchiolitis	Pediatric	3
19	Female	Bronchiolitis	Pediatric	3
20	Female	Common Cold	Adult	3

Demographics and Diagnoses

Which diagnoses are most common?

```
SELECT TOP 5
diagnosis,
COUNT(*) AS total_patient_count
FROM [Healthcare_Database].[dbo].[Outpatient Visits]
WHERE diagnosis <> 'Unknown'
GROUP BY diagnosis
ORDER BY total_patient_count DESC
```

	diagnosis	total_patient_count
1	Diabetes	37
2	Respiratory Illness	26
3	Muscle Injury	24
4	Hypertension	23
5	Common Cold	13

Appointments

When do appointments occur most often across the day?

```
SELECT
DATEPART (hour, appointment_time) AS appointment_hour,
COUNT(*) AS appointment_count
FROM [Healthcare_Database].[dbo].Appointments
GROUP BY
DATEPART (hour, appointment_time)
ORDER BY appointment_count DESC
```

	appointment_hour	appointment_count
1	12	46
2	9	40
3	11	38
4	10	34
5	8	16

Labs

Which lab tests are ordered most frequently?

```
SELECT
test_name,
COUNT(*) AS test_count
FROM [Healthcare_Database].[dbo].[Lab Results]
GROUP BY test_name
ORDER BY test_count DESC
```

	test_name	test_count
1	Chloride	49
2	Fasting Blood Sugar	42
3	ALT	40
4	Thyroid Stimulating Hormone	40
5	White Blood Cells	39
6	Uric Acid	37
7	Hemoglobin A1C	35
8	C-Reactive Protein	35
9	HDL Cholesterol	34
10	Creatinine	29

Labs

Which patients have fasting blood sugar results outside the normal range of 70–100 mg/dL?

```
SELECT
p.patient_id,
p.patient_name,
result_value
FROM [Healthcare_Database].[dbo].Patients AS p
INNER JOIN [Healthcare_Database].[dbo].[Outpatient Visits] AS ov
ON p.patient_id = ov.patient_id
INNER JOIN [Healthcare_Database].[dbo].[Lab Results] AS lr
ON ov.visit_id = lr.visit_id
WHERE lr.test_name = 'Fasting Blood Sugar'
AND (lr.result_value < 70 OR lr.result_value >100)
```

	patient_id	patient_name	result_value
1	521001	Emma Johnson	107
2	521001	Emma Johnson	101
3	521002	Michael Smith	103
4	521004	William Jones	110
5	521004	William Jones	105.82
6	521008	Bethany Clark	58.95
7	521009	Jose Gonzalez	110
8	521011	Amelia Thomas	111.47
9	521011	Amelia Thomas	112
10	521013	Alexander Perez	55.87
11	521021	Olivia Scott	107.99
12	521024	Liam Baker	124.81
13	521030	James Cooper	66.38
14	521031	Andy Morris	127
15	521032	Rick Sanders	50.91
16	521033	Eva Torres	123.27
17	521047	Nick Reid	55.34
18	521069	Sara Martin	120
19	521074	Sophie Myers	105
20	521092	Ava White	110

Risk Stratification

How many patients fall into each risk category?

- High risk: Smokers diagnosed with either hypertension or diabetes.
- Medium risk: Non-smokers diagnosed with either hypertension or diabetes.
- Low risk: Patients who do not meet the criteria above.

```sql
SELECT
CASE
WHEN smoker_status = 'Y' AND (diagnosis = 'Hypertension' OR diagnosis = 'Diabetes') THEN 'High Risk'
WHEN smoker_status = 'N' AND (diagnosis = 'Hypertension' OR diagnosis = 'Diabetes') THEN 'Medium Risk'
ELSE 'Low Risk'
END AS Risk_category,
COUNT(patient_id) AS num_patients
FROM [Healthcare_Database].[dbo].[Outpatient Visits]
GROUP BY
CASE
WHEN smoker_status = 'Y' AND (diagnosis = 'Hypertension' OR diagnosis = 'Diabetes') THEN 'High Risk'
WHEN smoker_status = 'N' AND (diagnosis = 'Hypertension' OR diagnosis = 'Diabetes') THEN 'Medium Risk'
ELSE 'Low Risk'
END
```

	Risk_category	num_patients
1	High Risk	21
2	Low Risk	389
3	Medium Risk	39

Readmissions

Which patients returned for another visit within 30 days?

```sql
SELECT
ov_initial.patient_id,
ov_initial.visit_date AS initial_visit_date,
ov_initial.reason_for_visit AS reason_for_initial_visit,
ov_readmit.visit_date AS readmission_date,
ov_readmit.reason_for_visit AS reason_for_readmission,
DATEDIFF(day, ov_initial.visit_date, ov_readmit.visit_date) AS days_between
FROM
[Healthcare_database].[dbo].[Outpatient Visits] AS ov_initial
INNER JOIN [Healthcare_database].[dbo].[Outpatient Visits] AS ov_readmit
ON ov_initial.patient_id = ov_readmit.patient_id
WHERE DATEDIFF(day, ov_initial.visit_date, ov_readmit.visit_date) <= 30
AND ov_readmit.visit_date > ov_initial.visit_date
```

See next page for detailed output.

Detailed Output

	patient_id	initial_visit_date	reason_for_initial_visit	readmission_date	reason_for_readmission	days_between
1	521025	2024-12-07 00:00:00.000	Checkup	2024-12-31 00:00:00.000	Checkup	24
2	521026	2024-05-31 00:00:00.000	Knee pain	2024-06-10 00:00:00.000	Checkup	10
3	521027	2024-10-10 00:00:00.000	Diabetes check	2024-11-08 00:00:00.000	Annual physical	29
4	521027	2024-11-08 00:00:00.000	Annual physical	2024-11-10 00:00:00.000	Diet and Exercise Counseling	2
5	521028	2023-05-29 00:00:00.000	Fever	2023-06-15 00:00:00.000	Annual physical	17
6	521029	2024-01-19 00:00:00.000	Annual physical	2024-01-30 00:00:00.000	Diet and Exercise Counseling	11
7	521029	2024-01-19 00:00:00.000	Annual physical	2024-02-01 00:00:00.000	Diabetes check	13
8	521029	2024-01-30 00:00:00.000	Diet and Exercise Counseling	2024-02-01 00:00:00.000	Diabetes check	2
9	521030	2023-02-11 00:00:00.000	Checkup	2023-02-25 00:00:00.000	Checkup	14
10	521030	2024-10-30 00:00:00.000	Annual physical	2024-11-14 00:00:00.000	Diabetes check	15
11	521032	2024-03-10 00:00:00.000	Fever	2024-03-15 00:00:00.000	Headache	5
12	521032	2024-03-10 00:00:00.000	Fever	2024-03-30 00:00:00.000	Back pain	20
13	521032	2024-03-15 00:00:00.000	Headache	2024-03-30 00:00:00.000	Back pain	15
14	521033	2024-08-08 00:00:00.000	Knee pain	2024-08-27 00:00:00.000	Fever	19
15	521034	2024-08-01 00:00:00.000	Diabetes check	2024-08-16 00:00:00.000	Annual physical	15
16	521034	2024-11-10 00:00:00.000	Checkup	2024-11-23 00:00:00.000	Checkup	13
17	521034	2024-11-10 00:00:00.000	Checkup	2024-12-07 00:00:00.000	Back pain	27
18	521034	2024-11-23 00:00:00.000	Checkup	2024-12-07 00:00:00.000	Back pain	14
19	521035	2023-04-20 00:00:00.000	Back pain	2023-05-17 00:00:00.000	Hypertension	27
20	521035	2023-07-02 00:00:00.000	Headache	2023-07-23 00:00:00.000	Fever	21